

INDUSTRIAL TRAINING REPORT

on

“Automated Resource Alert & Notification System”

by

Sameer Verma [Roll No.- 2410014045066]

**Submitted on Partial Fulfillment of the Requirements for the award of
the Degree
of**

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“Google Career Certificates”



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DECLARATION

I, **Sameer Verma**, hereby declare that the Summer Internship Report titled “**Automated Resource Alert & Notification System**” is an original work carried out by me during my training period. This report has not been submitted to any other institution or university for the award of any degree or diploma.

I have completed this internship/training under Google IT Automation with Python, as certified by the Google Career Certificates programme. The training provided me with practical and hands-on expertise in Python automation, system administration tasks, debugging, and process optimization.

I further declare that this report is a genuine representation of the work accomplished during the training, and all information has been presented accurately to the best of my knowledge.

Date: 20th December 2025

Signature
(Sameer Verma)

CERTIFICATE



7 Courses

Crash Course on Python

Using Python to Interact
with the Operating System

Introduction to Git and
GitHub

Troubleshooting and
Debugging Techniques

Configuration Management
and the Cloud

Automating Real-World
Tasks with Python

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Jul 6, 2025

SAMEER VERMA

has successfully completed the online, non-credit Professional
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Google IT Automation with Python

This six-course certificate, developed by Google, is designed to provide IT professionals with in-demand skills -- including Python, Git, and IT automation -- that can help them advance their careers. The hands-on curriculum is designed to teach learners how to write code in Python, with a special focus on how this applies to automating tasks in the world of IT support and systems administration. Those who received this certificate passed all graded assessments with a score of 80% of above. They should have a strong foundation in how to use Git and GitHub, troubleshoot and debug complex problems, and apply automation at scale by using configuration management and the Cloud in order to prepare them for more advanced IT Support Specialist or Junior Systems Administrator positions.

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Global Director of
Google Career
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ABSTRACT

This project report presents the design and implementation of an Automated Resource Alert & Notification System, developed during my Summer Training under the Google IT Automation with Python program. Modern computer systems require continuous monitoring to ensure stable performance and prevent failures. The proposed automation script monitors essential system resources such as CPU usage, RAM consumption, and Disk utilization and automatically triggers alerts when usage exceeds predefined thresholds.

The project incorporates Python automation libraries such as psutil, smtplib, and mysql.connector to perform system monitoring, generate log records, and notify the support team through email. Whenever resource usage crosses the specified limit, the system captures the health status, updates the alert log in the database, and sends real-time notifications to the technical team.

This automated solution reduces manual effort, improves reliability, and enables proactive issue management in environments where system performance is critical. The report also highlights the technical skills gained during the Google IT Automation course and demonstrates the application of Python programming in solving real-world IT operations challenges.

ACKNOWLEDGEMENT

I would like to express my sincere and heartfelt gratitude and appreciation to all the individuals who helped me in successfully completing this project on “**Automated Resource Alert & Notification System**” within the limited time frame.

I express my sincere gratitude to **Google Career Certificates** for offering the *Google IT Automation with Python* program, which served as the foundation for my Summer Internship project. The hands-on learning modules, practical assignments, and real-world automation exercises immensely contributed to shaping my technical understanding of Python, Git, system administration, and IT automation

.
I would also like to thank my faculty members and the Department of Computer Applications for providing continuous guidance, motivation, and support throughout the preparation of this report. Their valuable insights encouraged me to explore automation concepts in greater depth.

I would also like to extend my sincere thanks to all my classmates and peers for their collaboration, and the positive learning environment along with healthy criticism. The journey of working on this project was made more meaningful through the shared experiences, discussions, and mutual support

.
Lastly, I express my gratitude to **Team Coursera** for providing the necessary resources and an academic environment conducive to learning and growth. This project has been an invaluable part of my educational journey, and I am truly grateful for everyone who contributed to its success.

Thank you all for making this assignment a fulfilling and rewarding experience.

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INTRODUCTION

In the rapidly evolving technological landscape, maintaining system stability and performance has become essential for ensuring uninterrupted digital operations. System resources such as **CPU, RAM, and Disk Storage** play a crucial role in determining the efficiency and responsiveness of applications. When these resources exceed acceptable usage thresholds, system slowdowns or failures may occur, ultimately impacting user experience and business processes.

During my Summer Training under the **Google IT Automation with Python Certificate Program**, I learned the importance of automating repetitive administrative tasks to enhance efficiency and minimize error. The program equipped me with the skills needed to design solutions that automate monitoring, debugging, reporting, and notification workflows commonly used in IT support and system administration.

This project, **Automated Resource Alert & Notification System**, reflects the practical application of the core concepts of IT automation. The system performs an automated resource check at the moment of execution, identifies performance bottlenecks, and automatically alerts the support team via email. This reduces manual supervision, decreases downtime, and supports proactive system maintenance.

The purpose of this report is to document the training objectives, the technology used, the complete system workflow, and the program logic that enables automated monitoring. Through this project, I have gained hands-on experience in Python scripting, networking essentials, email automation, database connectivity, and error-handling techniques.

OVERVIEW OF COMPANY

The project was completed as part of the Google IT Automation with Python Professional Certificate, offered through Coursera. The certificate consists of six comprehensive courses that provide learners with practical, industry-relevant knowledge.

Google's internal IT infrastructure manages millions of servers, data centers, cloud workloads, and global services. To operate such systems efficiently, Google developed advanced automation practices that reduce manual work and improve system reliability.

The **Google Career Certificates** program was created to make these industry-level skills accessible to learners worldwide, bridging the gap between academic knowledge and real-world technical job skills.

Coursera offers a wide range of courses in fields such as computer science, information technology, data science, business, artificial intelligence, and programming languages. Many of these courses are designed and authorized by reputed organizations such as Google, IBM, Meta, Microsoft, and top universities worldwide. The platform follows a structured learning approach that includes video lectures, reading materials, quizzes, assignments, and practical exercises.

One of the major advantages of Coursera is its flexible learning model, which allows students to learn at their own pace. This makes it particularly suitable for students who wish to utilize their summer vacation productively to acquire new skills and enhance their technical knowledge. Coursera also provides verified certificates upon successful completion of courses, which can be used as proof of training and skill acquisition.

Key Skills Acquired through this program:

- Python scripting
- OS automation
- System monitoring
- Error/debug handling
- Real-world task automation
- Cloud & configuration concepts

OVERVIEW OF TECHNOLOGY

The project integrates multiple technologies, each serving a specific purpose:

1. Python Programming Language:

Python was chosen due to its readability, extensive libraries, and strong support for automation.

Key modules used include:

- **psutil** – for reading system resource usage (CPU, RAM, Disk).
- **smtplib** – to send emails using Gmail SMTP protocol.
- **email.message** – for constructing rich email messages.
- **mysql.connector** – for storing and retrieving alert logs and contact info.
-

2. MySQL Database:

The MySQL database stores:

- Support team details (names, email IDs)
- Alert logs with timestamp, resource type, message

This enables record-keeping and auditing of system warnings.

3. SMTP (Simple Mail Transfer Protocol):

Used for sending automated alerts via:

- Gmail SMTP Server
- TLS encryption for secure transmission
-

4. System Monitoring Technology:

Using psutil, the system captures:

- CPU Utilization
- RAM Usage
- Disk Partition Status

Thresholds determine when an alert needs to be generated.

PROJECT DETAILS

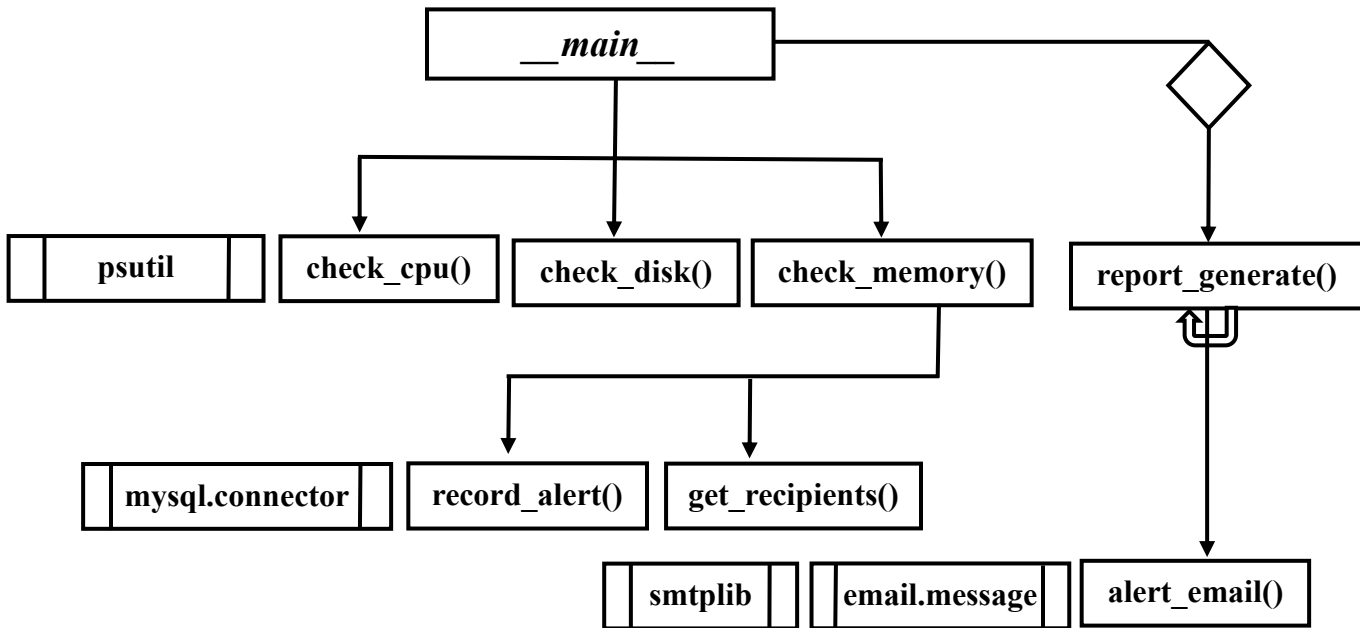
Project Objective:

To develop an automated system that:

- On demand monitors system resources
- Detects abnormal or high usage
- Sends real-time email alerts to the support team
- Logs all alerts in the database

Functional Workflow:

1. Check CPU usage
2. Check RAM usage
3. Check Disk usage (each partition)
4. Compare with thresholds
5. If exceeded:
 - Update MySQL alert log
 - Fetch support team emails
 - Send personalized email notifications
6. Display overall system health



Source Code

MySQL (Database Structure):

- CREATE TABLE tech_members (

First_Name VARCHAR(50),
 Last_Name VARCHAR(50),
 Email VARCHAR(100) PRIMARY KEY);
- INSERT INTO tech_members (first_name, last_name, email) VALUES

('Sameer', 'Verma', 'sameer81810@gmail.com'),
 ('Arun', 'Verma', 'arunverma9415@gmail.com'),
 ('Harshita', 'Verma', 'harshitavema1124@gmail.com')

First_Name	Last_Name	Email
Arun	Verma	arunverma9415@gmail.com
Harshita	Verma	harshitavema1124@gmail.com
Sameer	Verma	sameer81810@gmail.com

➤ CREATE TABLE Alert_Log (
 Log_Time DATETIME PRIMARY KEY,
 Type VARCHAR(50),
 Error_Msg VARCHAR(100));

Log_Time	Type	Error_Msg
2025-12-05 16:36:08	RAM	RAM Limit Reached.

Python Script:

```
import psutil
import mysql.connector
import smtplib
from email.message import EmailMessage
flag={"CPU":0,"Disk":0,"RAM":0}

def check_cpu():
    usage=psutil.cpu_percent(interval=1)
    #compare CPU usage level
    if usage>80:
        flag["CPU"]=1
    return usage

def check_disk():
    temp=[]
    for path in psutil.disk_partitions(all=False):
        p=path.mountpoint
        #Disconnected pendrive/external ssd can raise error
        try:
            stat=psutil.disk_usage(p)
            usage=stat.percent
```

```

        except (PermissionError, OSError):
            continue
        temp.append((p,usage))
        #compare disk usage level of particular partition
        if usage>75:
            flag["Disk"]=1
            break
    return temp

```

```

def check_memory():
    stat=psutil.virtual_memory()
    usage=stat.percent
    #compare RAM usage level
    if usage>80:
        flag["RAM"]=1
    return usage

```

```

def get_recipients():
    conn=mysql.connector.connect(host='localhost',user='root',passwd='1234',
                                database='Support_Team')

    if conn.is_connected()== False:
        print("Unable To Connect To Support Team...")
        #returning empty list if failed to fetch detail
        return [],[]
    cur=conn.cursor()
    cur.execute("SELECT * FROM tech_members")
    results = cur.fetchall()
    conn.close()          #Column 1-> First Name
    Names=[]              #Column 2-> Last Name
    Emails=[]             #Column 3-> Email Address
    for x in results:
        Names.append(x[0]+' '+x[1])
        Emails.append(x[2])
    return Names,Emails

```

```

def record_alert(Type,Msg):
    conn=mysql.connector.connect(host='localhost',user='root',passwd='1234',
                                database='Support_Team')
    if conn.is_connected()== False:
        #Raising error for missing record log and return
        print("Alert Cannot Be Recorded To Database...")
        return
    cur=conn.cursor()
    cur.execute("INSERT INTO Alert_Log VALUES (NOW(),%s,%s)",(Type,Msg))
    conn.commit()
    print("Alert Raised")
    print("WARNING: ",Msg)
    conn.close()

```

```

def alert_email(subject,body,email):
    # Gmail SMTP server details
    SMTP_SERVER = "smtp.gmail.com"
    SMTP_PORT = 587
    #Email credentials
    sender="sameer81810@gmail.com"
    reciever=email
    password="*****"
    # Create the email
    msg = EmailMessage()
    msg["From"] =sender
    msg["To"] = reciever
    msg["Subject"] = subject
    msg.set_content(body)
    try:
        # Connect to Gmail SMTP server
        server = smtplib.SMTP(SMTP_SERVER, SMTP_PORT)
        server.starttls() # Secure the connection
        server.login(sender, password)
        server.send_message(msg)

```

```

        server.quit()
        print("To:",receiver,"\tEmail sent successfully!")
except Exception as e:
    print("Failed to send email:", e)

```

```

def report_generate(cpu,p,disk,ram):
    res=[]
    text1=text2=text3=""
    #check which resource limit exceed
    if flag["CPU"]:
        res.append("CPU")
        text1=f"C.P.U. Usage: {cpu}%. \n"
    if flag["Disk"]:
        res.append("Disk")
        text2=f"Disk Partition {p} Usage: {disk}%. \n"
    if flag["RAM"]:
        res.append("RAM")
        text3=f"R.A.M. Usage: {ram}%. \n"

    res=", ".join(res)
    subject="{ } Limit Reached.".format(res)
    #Database log update
    record_alert(res,subject)
    #Fetch support team details
    names,emails=get_recipients()
    #set mail body for individual
    for i in range(len(names)):
        body=""
        body+=f"\n_____RESOURCE ALERT_____ \n\nDear { },\n"
        body+=f"Following resources usage has reached its limit.\n"
        body+=f"\t{ } \t{ } \t{ } \nPlease check active processes and available"
        body+=f"hardware to ensure flawless execution & smooth user"
        body+=f"experience.".format(names[i],text1,text2,text3)
        #calling function to send mail
        alert_email(subject,body,emails[i]

```

```
#calling resource monitor functions
cpu=check_cpu()
disks=check_disk()
ram=check_memory()
#check flags to ensure if alert needed
for x in flag.values():
    if x != 0:
        health=1
        break
    else:
        health=0
if health:
    #generate alert for unhealthy state
    print("-SYSTEM IS IN UNHEALTHY STATE-\n")
    p,disk=disks[-1]
    report_generate(cpu,p,disk,ram)
else:
    print("-SYSTEM IS IN HEALTHY STATE-\n")
print("C.P.U. Usage->",cpu)
for p,disk in disks:
    print("Disk Usage","for",p,"->",disk)
print("RAM Usage->",ram)
```

Source Code Explanation

- **check_cpu()**
Uses psutil.cpu_percent() to capture current CPU load. Marks a flag if > 80%.
- **check_memory()**
Reads RAM utilization using psutil.virtual_memory().
- **check_disk()**
Iterates through all mounted partitions and checks usage. Flags when disk > 75%.
- **get_recipients()**
Connects to MySQL database and retrieves support team names & emails.
- **record_alert()**
Inserts an alert entry with timestamp into the MySQL Alert_Log table.
- **alert_email()**
Sends formatted alert emails using SMTP server.
- **report_generate()**
Prepares the alert body text and sends notifications to all registered members.
- **Main Script**
Executes checks, determines health, triggers alerts, prints summary.

OUTPUT

Under Threshold:

```
-SYSTEM IS IN HEALTHY STATE-  
  
C.P.U. Usage-> 5.4  
Disk Usage for C:\ -> 22.0  
Disk Usage for E:\ -> 10.9  
Disk Usage for F:\ -> 23.8  
RAM Usage-> 75.1
```

Over Threshold:

-SYSTEM IS IN UNHEALTHY STATE-

Alert Raised

WARNING: CPU, RAM Limit Reached.

To: arunverma9415@gmail.com [Email sent successfully!]

To: harshitaverma1124@gmail.com [Email sent successfully!]

To: sameer81810@gmail.com [Email sent successfully!]

C.P.U. Usage-> 100.0

Disk Usage for C:\ -> 22.0

Disk Usage for E:\ -> 10.9

Disk Usage for F:\ -> 23.8

RAM Usage-> 86.8

Received E-Mail



CPU, RAM Limit Reached.

1 message

<sameer81810@gmail.com>

Sat, Jan 3, 2026 at 8:35 PM

To: arunverma9415@gmail.com

____RESOURCE ALERT____

Dear Arun Verma,

Following resources usage has reached its limit.

C.P.U. Usage: 100.0%.

R.A.M. Usage: 86.8%.

Please check active processes and available hardware to ensure flawless execution & smooth user experience.

Updated Alert_Log

Log_Time	Type	Error_Msg
2025-12-05 16:36:08	RAM	RAM Limit Reached.
2026-01-03 19:00:29	RAM	RAM Limit Reached.
2026-01-03 20:35:25	CPU, RAM	CPU, RAM Limit Reached.

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